

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	6aba0fe75015c35b98630d90
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The architecture below describes a service-management solution in which employees submit standard laptop orders through a service catalog, Flow Designer automates validation and approval, and procurement and asset records are updated as the order moves through fulfillment.

Logical Architecture – Standard Laptop Order Automation

Employee	→	Service Catalog / Portal	→	Laptop Request
Laptop Request	→	Flow Designer	→	Validation
Validation	→	Approval Workflow	→	Approver
Approved Request	→	Procurement Integration	→	Procurement Record / Order
Order	→	Vendor / Fulfillment Interface	→	Shipment / Delivery
Delivery	→	Asset Management	→	Assigned Laptop Asset
Workflow Status	←	Flow Designer	←	Employee Notifications

Architecture considerations:

- Application logic is implemented through Flow Designer workflows for validation, approval, routing, notifications, and task creation.
- The service catalog and portal form the user-facing interface.
- Procurement and asset-management records provide persistent operational data.
- External vendor/procurement interfaces are represented as integration points where applicable.
- The solution is intended to be deployed within the organization's ServiceNow/cloud environment rather than as a standalone local application.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employee interaction with the application through the service catalog / self-service portal.	ServiceNow Service Catalog, Service Portal / Employee Center
2	Application Logic-1	Validate standard laptop selection, required fields and request eligibility.	ServiceNow Flow Designer
3	Application Logic-2	Automate approval routing, task creation, notifications and workflow state changes.	Flow Designer Actions, Approvals and Notifications
4	Application Logic-3	Manage procurement and asset lifecycle activities associated with the approved request.	ServiceNow Procurement / Asset Management capabilities
5	Database	Store requests, approvals, tasks, users, order status and related operational records.	ServiceNow platform database
6	Cloud Database	Cloud-hosted persistence for the ServiceNow application and its platform records.	ServiceNow cloud data platform
7	File Storage	Optional storage for procurement documents or supporting attachments.	ServiceNow attachment storage / configured cloud storage
8	External API-1	Interface with procurement or vendor systems to place orders or obtain order status, if required.	REST API / IntegrationHub connector
9	External API-2	Optional interface for shipping or delivery tracking information.	REST API / Vendor or carrier integration
10	Machine Learning Model	No machine-learning model is required for the standard automated ordering workflow.	Not applicable
11	Infrastructure (Server / Cloud)	Application deployment and workflow execution environment.	ServiceNow Cloud / SaaS; local infrastructure not required

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core workflow; the solution uses the ServiceNow platform capabilities.	ServiceNow Flow Designer / platform services
2	Security Implementations	Role-based access, authentication, authorization, controlled catalog access, auditability and secure integration endpoints should be applied.	ServiceNow IAM / roles, ACLs, authentication, HTTPS/TLS, audit logs
3	Scalable Architecture	Workflow components are modular and can support additional laptop models, approval rules, departments and procurement paths without redesigning the complete solution.	Flow Designer, reusable actions/subflows, ServiceNow platform architecture
4	Availability	The solution relies on the availability of the ServiceNow cloud platform and configured integration endpoints; workflow state is retained in platform records.	ServiceNow cloud availability, monitored integrations, retry/error handling
5	Performance	Standard laptop requests should use automated validation and workflow actions to reduce manual processing. Integration calls should use efficient APIs and avoid unnecessary synchronous steps.	Flow Designer, IntegrationHub/REST, indexed platform records, asynchronous processing where appropriate

References:

- ServiceNow platform documentation and Flow Designer documentation.
- ServiceNow Service Catalog and Asset Management documentation.
- ServiceNow IntegrationHub / REST integration documentation.